

DILLON 18 WSW, MT, USA

WMO: 742270

Lat: 45.1581N

Lon: 113.0057W

Elev: 1820

StdP: 81.29

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 04137

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.4	-23.3	-29.9	0.3	-26.8	-26.1	0.4	-23.1	8.6	-2.7	7.6	-2.3	0.8	N/A	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	30.1	12.2	28.9	12.0	27.2	11.7	14.0	25.3	13.2	24.8	12.6	24.3	4.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	9.6	16.2	8.7	8.7	15.7	7.6	8.1	15.1	45.2	25.2	42.8	24.8	40.8	24.4	18.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.3	DB	-33.6	32.7	3.7	0.9	-36.3	33.3	-38.4	33.8	-40.5	34.3	-43.2	34.9	(4)
(5)			WB	-33.5	15.8	4.3	1.2	-36.6	16.7	-39.1	17.4	-41.5	18.1	-44.6	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-7.4	-6.5	-1.1	3.0	7.8	12.6	17.3	16.0	11.2	4.4	-2.7	-8.7	(6)	
	DBStd	10.06	6.31	6.38	5.18	3.98	3.82	3.74	2.44	2.62	3.92	4.62	6.06	6.44	(7)	
	HDD10.0	2837	539	461	344	212	88	18	0	1	32	181	381	581	(8)	
	HDD18.3	5295	797	694	603	460	326	173	48	78	213	433	631	839	(9)	
	CDD10.0	606	0	0	0	2	21	96	225	188	69	6	0	0	(10)	
	CDD18.3	23	0	0	0	0	0	2	14	6	1	0	0	0	(11)	
	CDH23.3	1586	0	0	0	0	10	170	728	539	134	5	0	0	(12)	
	CDH26.7	418	0	0	0	0	0	39	214	143	22	0	0	0	(13)	
	WSAvg	2.3	2.0	2.3	2.4	2.8	2.5	2.5	2.3	2.2	2.1	2.2	2.0	1.9	(14)	
(15) Precipitation	PrecAvg	284	14	10	16	28	40	52	26	23	20	23	18	14	(15)	
	PrecMax	431	36	25	38	45	68	87	90	49	44	55	51	29	(16)	
	PrecMin	202	2	1	4	8	17	14	0	6	1	1	3	6	(17)	
	PrecStd	60	9	6	8	9	15	24	20	14	12	15	12	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	9.1	15.8	20.4	24.8	30.1	31.6	31.7	29.4	23.1	14.5	6.9	(19)	
		MCWB	1.5	2.5	5.2	7.5	9.6	12.0	12.9	12.1	11.5	10.1	6.1	1.9	(20)	
	2%	DB	4.5	5.8	11.6	17.8	22.3	27.7	30.3	29.9	26.6	18.8	11.5	4.6	(21)	
		MCWB	0.7	1.6	4.0	6.7	9.3	12.0	12.3	12.0	10.6	7.8	4.7	1.0	(22)	
	5%	DB	3.0	3.8	8.9	14.9	19.8	25.2	29.2	28.4	24.8	16.3	8.7	2.6	(23)	
		MCWB	-0.3	0.4	2.9	5.3	8.6	11.3	12.2	11.5	10.1	7.0	3.1	-0.4	(24)	
	10%	DB	1.2	2.3	6.6	11.7	17.1	22.7	27.7	26.6	22.5	13.4	6.3	0.5	(25)	
		MCWB	-1.5	-0.7	1.8	4.0	7.7	10.6	12.0	11.2	9.6	5.8	1.8	-2.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.2	6.5	8.9	11.5	14.8	15.5	14.4	14.0	10.4	6.7	2.7	(27)	
		MCDB	5.8	8.3	13.9	18.2	20.0	25.9	26.7	26.0	21.0	20.8	13.9	5.9	(28)	
	2%	WB	1.2	2.0	4.8	7.0	10.2	13.3	14.2	13.3	12.1	9.1	5.0	1.6	(29)	
		MCDB	4.4	5.6	11.2	16.3	18.7	25.1	25.3	24.8	22.7	17.2	10.5	4.3	(30)	
	5%	WB	0.1	0.8	3.6	5.8	9.2	12.2	13.5	12.6	11.0	7.7	3.6	-0.1	(31)	
		MCDB	3.1	4.0	8.8	14.0	17.7	22.6	25.3	24.5	22.4	14.9	8.5	2.5	(32)	
	10%	WB	-1.1	-0.2	2.3	4.5	8.3	11.3	12.8	11.9	10.1	6.4	1.8	-1.8	(33)	
		MCDB	1.5	2.7	6.9	11.4	16.2	21.5	25.0	24.4	21.1	13.2	5.7	0.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.1	11.5	13.0	13.6	15.7	19.4	19.5	17.7	14.1	12.6	11.6	(35)	
		MCDBR	11.6	9.9	14.7	19.6	19.7	21.2	22.1	23.1	23.3	20.1	16.3	11.3	(36)	
	5% WB	MCWBR	8.6	6.9	8.1	10.0	8.9	8.6	8.1	8.9	10.2	10.3	10.1	8.4	(37)	
		MCWBR	11.1	9.7	13.7	18.0	16.4	19.0	19.7	19.9	20.2	17.7	15.4	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.229	0.238	0.255	0.292	0.300	0.306	0.312	0.316	0.292	0.268	0.246	0.229	(40)	
		taud	2.417	2.391	2.476	2.447	2.514	2.534	2.541	2.491	2.560	2.598	2.563	2.474	(41)	
	Edn at Noon	Edn at Noon	917	969	993	972	968	960	949	935	939	925	895	885	(42)	
		Edh at Noon	69	87	93	105	102	101	99	100	85	71	60	59	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.64	2.62	3.92	5.09	5.82	6.75	7.14	6.16	4.71	3.03	1.84	1.33	(44)	
	RadStd	RadStd	0.16	0.23	0.24	0.37	0.55	0.54	0.38	0.34	0.39	0.24	0.14	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.66	N/A	N/A	N/A	N/A	

Nomenclature: See separate page